Inventory Management System

Simple C++ console app to manage stock items in a store

Source Code (inventory.cpp)

```
// Inventory Management System
// Simple console app to manage stock items in a store

#include <iostream>
#include <vector>
#include <string>
using namespace std;

struct Item {
    int id;
    string name;
    int quantity;
    double price;
};

vector<Item> inventory;
int nextId = 1;

void addItem() {
    Item item;
    item.id = nextId++;

    cout << "Enter item name: ";
    cin.ignore();
    getline(cin, item.name);

    cout << "Enter quantity: ";
    cin >> item.quantity;

    cout << "Enter price per unit: ";
    cin >> item.price;

    inventory.push_back(item);
    cout << "Item added with ID " << item.id << endl;
}

void displayItems() {
    if (inventory.empty()) {
        cout << "Inventory is empty." << endl;
        return;
    }

    cout << "\nID\tName\t\tQty\tPrice\tTotal" << endl;
    cout << "-----------------------------------------" << endl;
    for (const Item &item : inventory) {
        double total = item.quantity * item.price;
        cout << item.id << "\t" << item.name << "\t\t" << item.quantity
             << "\t" << item.price << "\t" << total << endl;
    }
}

void updateStock() {
    int id, qty;
```

```
    cout << "Enter item ID to update: ";
    cin >> id;

    for (Item &item : inventory) {
        if (item.id == id) {
            cout << "Enter quantity to add (use negative to remove): ";
            cin >> qty;

            if (item.quantity + qty < 0) {
                cout << "Error: Not enough stock to remove that quantity." << endl;
                return;
            }

            item.quantity += qty;
            cout << "Stock updated. New quantity: " << item.quantity << endl;
            return;
        }
    }
    cout << "Item with ID " << id << " not found." << endl;
}

int main() {
    int choice;

    do {
        cout << "\n===== Inventory Management =====" << endl;
        cout << "1. Add Item" << endl;
        cout << "2. Display Items" << endl;
        cout << "3. Update Stock" << endl;
        cout << "4. Exit" << endl;
        cout << "Enter choice: ";
        cin >> choice;

        switch (choice) {
            case 1:
                addItem();
                break;
            case 2:
                displayItems();
                break;
            case 3:
                updateStock();
                break;
            case 4:
                cout << "Exiting program." << endl;
                break;
            default:
                cout << "Invalid choice, try again." << endl;
        }
    } while (choice != 4);

    return 0;
}
```